

Complements of Operational Research

7.1 Inventory management: ABC classification and economic order quantity

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Why hold stock — and why it costs

A stock is a buffer between the supplier and the consumer: it allows buying in batches, absorbing delays and meeting demand at once.

The trade-off

- order **little and often**: many ordering costs (transport, receiving, paperwork);
- order **a lot and rarely**: much idle capital, warehouse space, insurance, obsolescence.

Questions

1. Which items deserve attention? (**ABC**)
2. How much to order each time? (**EOQ**)
3. When to order? (**reorder point**)
4. What if we run out, if there are discounts, or if we produce it ourselves? (7.2 and 7.3)

The example in this chapter: a **paint factory**.

ABC classification: value is concentrated

Item	€/unit	qty/year	value	%	cum.	cl.
P1 titanium dioxide	4.00	12000	48000	39.9	39.9	A
P2 acrylic resin	3.20	9000	28800	23.9	63.8	A
P3 5 L cans	0.90	20000	18000	15.0	78.8	A
P4 organic pigments	25.00	400	10000	8.3	87.1	B
P5 solvent	1.10	6000	6600	5.5	92.6	B
P6 additives	12.00	300	3600	3.0	95.6	C
P7 calcium carbonate	0.15	18000	2700	2.2	97.8	C
P8 labels	0.03	40000	1200	1.0	98.8	C
P9 lids	0.05	22000	1100	0.9	99.8	C
P10 pallets	3.00	100	300	0.2	100	C

Total: 120 300 €/year. Items are sorted by **annual value** (price × quantity).

Rule (the one used in class)

A: up to 80% of the cumulative value;
B: up to 95%; C: the rest.

- **A**: 3 items (30%), 78.8% of the value: tight control, EOQ computed.
- **B**: 2 items, 13.8%.
- **C**: 5 items (50%), 7.4%: simple rules.

The pigments (25 €/kg) are B and the cans (0.90 €) are A: what counts is the annual value, not the price.

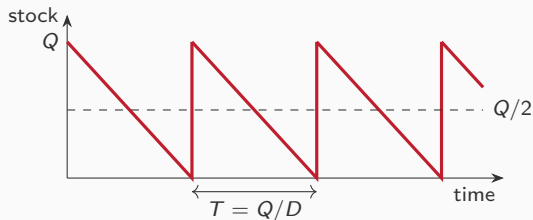
Wilson's model (EOQ)

Most important A item: titanium dioxide (P1).

- demand $D = 12\,000$ kg/year, constant;
- price $C_a = 4$ €/kg;
- cost per order $C_e = 150$ €;
- holding rate 20%/year $\Rightarrow$
 $C_p = 0.2 \times 4 = 0.80$ €/kg/year;
- lead time 2 weeks; no shortages.

Ordering Q each time: D/Q orders per year;
average stock $Q/2$.

$$K(Q) = D C_a + \frac{D}{Q} C_e + \frac{Q}{2} C_p$$

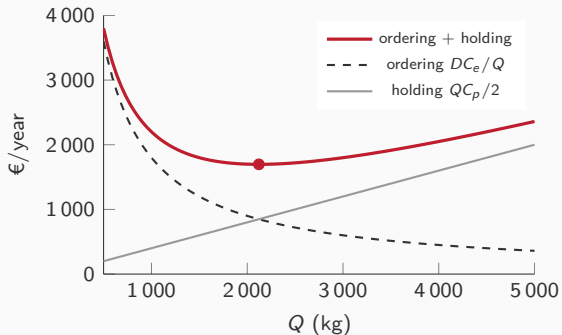


Optimum (differentiate and set to zero)

$$Q^* = \sqrt{\frac{2 D C_e}{C_p}}$$

At the optimum, ordering cost = holding cost.

Result for titanium dioxide



$Q^* = \sqrt{2 \cdot 12000 \cdot 150 / 0.8}$	2121 kg
orders per year D/Q^*	5.66
cycle $T = Q^*/D$	9.2 weeks
ordering + holding cost	1697.06 €
total cost (with purchase)	49 697.06 €
reorder point $D \times 2 \text{ wk}$	461.5 kg

Policy: when the stock falls to 462 kg, order 2121 kg (about one order every 9 weeks).

The EOQ is robust

Q (kg)	ordering + holding (€)	above optimum
1500	1800.00	+6.1%
2000	1700.00	+0.2%
2121	1697.06	—
2500	1720.00	+1.3%
3000	1800.00	+6.1%

Rounding to pallets of 2000 kg costs 3 €/year. A 40% error in Q costs about 6%.

What if demand was badly estimated?

If in fact $D = 14\,400$ kg (+20%) and we keep ordering 2121 kg, the cost is 1866.76 € against 1859.03 € at the optimum: **+0.4%**.

Why: Q^* depends on the *square root* of the data, and the cost curve is very flat near the minimum ($K/K^* = \frac{1}{2}(Q/Q^* + Q^*/Q)$).

The same flat bottom as in ch. 4: near the optimum, the decision tolerates errors in the data.

```
from math import sqrt

def abc(itens):
    # name -> (price, quantity)
    valor = {k: p * q for k, (p, q) in itens.items()}
    tot, acum = sum(valor.values()), 0
    for k in sorted(valor, key=valor.get, reverse=True):
        acum += valor[k]; c = 100 * acum / tot
        cl = 'A' if c <= 80 else 'B' if c <= 95 else 'C'
        print(k, round(c, 1), cl)

def qee(D, Ce, Cp, Ca=0):
    Q = sqrt(2 * D * Ce / Cp)
    return Q, D / Q, D * Ca + D * Ce / Q + Cp * Q / 2

print(qee(12000, 150, 0.8, 4)) # (2121.3, 5.66, 49697.06)
```

Full file (ABC, EOQ, shortages, discounts, production):
code/python/07_inventory/
07_1_abc_eoq/stocks.py (function names in Portuguese, as in the course code: qee = EOQ).

In a spreadsheet: sort by annual value, add a cumulative percentage column and an IF formula for the class.

Example 1 — ABC classification

Item	unit value	quantity used in a month
A1	1	22000
A2	12	410
A3	4.25	1468
A4	0.25	3500
A5	2.25	1600
A6	26	10
A7	8.5	124
A8	0.5	40000
A9	1.25	440
A10	0.12	25000

Example 2 — box of transistors

Price 15 m.u./unit; average demand 12 844 units/year; ordering cost 209 m.u.; holding cost 3.25 m.u./unit/year (insurance, warehouse, control and losses) plus 15% of the price (alternative return).

- Economic order quantity?
- Total inventory cost over one year?
- How often should orders be placed?
- With a lead time of 2 weeks, what is the reorder point?
- How many orders per year?
- Relation between holding and ordering costs in the optimal policy?

Exercise sheet 7 — Exercise 1 (homework)

ABC classification (homework; 2nd test, 27/06/2019)

Carry out an ABC classification of the products in the following table:

Product	A	B	C	D	E	F	G	H	I	J
Unit value	1.4	0.6	4.2	5	3	0.16	2.25	20	2.8	5
Quantity	400	5700	1300	600	2600	24000	3600	289	600	58

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 7](#).

KEY TAKEAWAYS — 7.1

1. ABC: sort by annual value; A up to 80% cumulative, B up to 95%, C the rest. Control A items carefully.
2. EOQ: $K(Q) = DC_a + \frac{D}{Q}C_e + \frac{Q}{2}C_p$; $Q^* = \sqrt{2DC_e/C_p}$; at the optimum, ordering = holding.
3. Cycle $T = Q^*/D$; number of orders D/Q^* ; reorder point = demand during the lead time.
4. The EOQ is robust: moderate errors in Q or in D cost very little.

Next: 7.2 What if shortages are allowed? What if there are discounts?

References and resources

References

Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Inventory Theory.
- Heizer, J., Render, B., Munson, C., *Operations Management: Sustainability and Supply Chain Management*, Pearson — ch. 12 (ABC, EOQ).
- Harris, F. W., How many parts to make at once, *Factory, The Magazine of Management*, 10(2), 135–136, 1913.

Resources

- Exercise sheet 7. `code/python/07_inventory/07_1_abc_eoq/stocks.py`.